Progress beyond the exponential benchmark

Status: 2026-09-12. This stage gives matching theory and polynomial procedures in additional regimes, rather than extending the small exponential optimizer. The general fixed-cap weak-signal computation problem remains open. All previous-stage files are preserved.

Main results

Requested direction	New conclusion	Remaining limit
Efficient high-dimensional minimax computation	If the smallest cap $r \gtrsim \sqrt{p}$, a structural/direct interval attains sharp length $\min(M\sqrt{r/p}, \tau)$ for every budget	Fixed and intermediate caps with diffuse weak signals remain open
Loading rates beyond small budgets	For fixed polynomial decay $\lvert w_i\rvert \le C\lvert w_1\rvert i^{-\beta}$, sharp length is $\lvert w_1\rvert \min(M, \tau)$ at every budget; a class of nonorthogonal perturbations is included	General profiles remain open; the matching block-fit construction is not an efficient optimizer
Growing-rank adaptation	Simultaneous minimax adaptation from growing outer caps to positive fixed inner caps is impossible	This is an impossibility for that objective, not a characterization of every adaptation opportunity
Pointwise inference	A polynomial cap-valid interval has optimal expected-length rate $\tau\sqrt{r/p}$ at strong diagonal Tucker means saturating their caps	Requires a conservative sufficient amplitude and norm slack; not optimal signal thresholds or leading constants
Computational limits	For power exponents $0 < \beta < 1/K$, a loading-compatible low-degree prior gives a polynomial gap from the new sharp estimation rate	Restricted polynomial-estimator result; no blanket hardness claim for efficient confidence intervals

The growing-cap conclusion uses a previously proved sparse-diagonal mixture and supplies its matching upper bound. The pointwise proof uses the same outer honesty class for both bounds and controls the pilot failure contribution with a budget shrinking as r/p. The loading proof uses dyadic blocks with explicit error allocations and a summable decay envelope. Complete supplied derivations are in theory/results.md and the supplement. These results have not undergone independent mathematical review.

Newly run growing-cap experiments

The interval reads only diagonal observations, so the experiment generates exactly those independent Gaussian coordinates. It does not materialize the unused off-diagonal noise. Coverage is proved for the full tensor class; simulation uses zero and rank-cap boundary means. The largest loading dimension is 16,384. The table shows its boundary cases; all dimensions and zero cases are retained in `results/growing_caps.json`.

p	Cap r	Structural radius / τ	Covered / trials	Coverage 95% Wilson interval	Mean length
16384	128	0.25	200/200	[0.981, 1.000]	0.500
16384	128	2.00	189/200	[0.904, 0.969]	3.920
16384	128	8.00	188/200	[0.898, 0.965]	3.920
16384	512	0.25	200/200	[0.981, 1.000]	0.500

p	Cap r	Structural radius / τ	Covered / trials	Coverage 95% Wilson interval	Mean length
16384	512	2.00	195/200	[0.943, 0.989]	3.920
16384	512	8.00	187/200	[0.892, 0.962]	3.920

There are 36 configurations and 7,200 newly run trials. The structural/direct choice is deterministic and the theoretical worst-case squared-risk bound is saved alongside empirical squared error. Finite Monte Carlo deviations are not used as proof of a uniform guarantee.

Newly run pointwise experiments

These experiments generate the complete tensor observation, then make a correctly scaled artificial split. No additional observed replicate is assumed. The supplied cap is retained. In the strong rows, the signal amplitude is eight times the theorem's conservative sufficient bound; it ranges from approximately 60,000 at $p = 8$ to 9.4 million at $p = 64$, with $\tau = 1$. These are deliberately very strong signals.

p	Cap r	Covered / trials	Coverage 95% Wilson interval	Mean length	Mean time (ms)
8	1	195/200	[0.943, 0.989]	2.672	0.37
8	3	193/200	[0.930, 0.983]	3.883	0.46
16	1	118/120	[0.941, 0.995]	1.931	0.96
16	4	116/120	[0.917, 0.987]	3.438	1.18
32	1	60/60	[0.940, 1.000]	1.430	6.28
32	6	58/60	[0.886, 0.991]	3.047	8.84
64	1	20/20	[0.839, 1.000]	1.003	74.75
64	8	19/20	[0.764, 0.991]	2.743	88.42

At rank one, the mean length decreases with dimension consistently with the proved $p^{-1/2}$ scale. All strong saturated trials choose the tangent option. Small sample counts at the largest dimensions produce wide coverage uncertainty; those rows are runtime/behavior checks, not high-precision coverage estimates.

The remaining cases test honesty outside the pointwise-optimality family:

Case	p	Cap	Covered / trials	Mean length	Chosen option
weak	8	1	484/500	1.889	structural: 500
strong_lower_rank	8	3	189/200	4.335	split_direct: 200
strong_plus_weak	8	3	193/200	4.315	split_direct: 200
dense_rank_one	16	1	197/200	2.081	structural: 200

The pointwise suite has 12 configurations and 1,900 new trials. Conservative fallback is retained when actual

rank is below the supplied cap or weak additional components are present. This is consistent with the limitation in the theorem, not a claim of full actual-rank adaptation.

Economy SVDs remove an avoidable allocation from the preceding prototype. For 64^3 input, the largest returned right singular matrix has 262,144 entries, compared with 16,777,216 for a full right singular matrix: a factor of 64 reduction in that array. This is an array-size comparison, not a measured factor-of-64 reduction in total memory or runtime.

Loading and adaptation calculations

There are 96 new loading-envelope calculations covering dimensions 64, 1,024 and 16,384, power exponents 0.1, 0.25, 0.5 and 1, orders three/four, and norm budgets from 0.1 to one million. They evaluate the deterministic diameter of ideal block fits. They do not numerically solve those fits or claim measured interval coverage. The fixed-decay proof is uniform over budgets and dimensions; its constants deteriorate as the decay exponent tends to zero.

Eight saved comparisons evaluate the growing-outer-cap lower bound against the positive fixed-cap oracle rate. The proved divergence rules out general simultaneous rate adaptation even for unlimited computation. The weighted low-degree proof separately explains why a sharp statistical loading rate need not yield a polynomial estimator.

Verification and remaining work

Fresh checks passed: 60 exact hypergeometric likelihood moments, 9 attained structural bounds, 32 dyadic/tail bounds, 60 pointwise bias/allocation cases, 50 independent projection comparisons, 9 weighted low-degree moment comparisons, and 4 signed unknown-support strong-signal checks. Seeds, actual software versions, source hashes, coverage intervals, timing, and the previous-stage manifest are saved.

Still open are efficient minimax intervals for the original fixed-cap diffuse weak-signal class, arbitrary loading geometry, intermediate growing-cap rates, a full pointwise characterization, and optimal signal thresholds for computation. The precise audit is in theory/audit.md; reproduction is in README.md.